

MATUA, Russian Federation

WMO: 322070

Lat: 48.051N

Lon: 153.255E

Elev: 58

StdP: 100.63

Time Zone: 11.00 (E11)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.7	-9.5	-16.3	0.9	-9.2	-15.0	1.0	-8.5	19.0	-6.0	17.5	-3.2	6.3	270	0.876

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.3	18.6	16.3	16.4	14.5	14.7	12.9	17.1	17.9	14.8	16.0	13.3	14.3	7.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.0	18.0	14.4	10.3	15.2	12.8	9.2	13.6	48.6	17.9	41.8	16.1	37.7	14.3	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	15.0	13.1	-13.0	21.2	1.8	2.7	-14.4	23.2	-15.4	24.8	-16.5	26.3	-17.8	28.3
			-13.9	18.7	1.6	2.4	-15.0	20.4	-16.0	21.9	-16.9	23.2	-18.1	25.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.7	-4.6	-4.2	-2.6	0.0	2.9	5.5	8.6	11.2	9.7	6.5	1.4	-2.0
	DBStd	5.81	2.66	2.88	2.13	1.82	2.06	2.51	2.86	2.52	1.99	1.98	2.44	2.47
	HDD10.0	2746	452	398	390	299	221	139	63	15	28	110	258	372
	HDD18.3	5693	711	631	648	549	478	386	303	222	259	367	508	630
	CDD10.0	95	0	0	0	0	1	3	19	52	19	1	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.7	7.0	7.1	7.5	7.0	5.2	3.8	3.6	4.0	4.7	5.8	6.4	6.6
Precipitation	PrecAvg	947	57	46	54	63	71	71	98	103	113	102	94	77
	PrecMax	1150	152	89	97	118	109	101	165	283	174	155	149	155
	PrecMin	727	22	18	29	17	24	20	31	12	56	55	44	35
	PrecStd	110	27	18	15	21	20	22	32	51	30	26	30	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	1.6	2.6	7.4	14.5	18.7	22.4	20.4	17.3	13.1	8.0	4.9
		MCWB	0.8	0.7	2.0	4.3	9.0	13.2	19.7	18.0	15.3	11.1	7.2	4.2
	2%	DB	0.9	0.8	1.1	4.2	9.7	14.1	18.9	18.5	15.4	11.3	6.9	3.5
		MCWB	0.1	0.0	0.3	1.8	6.2	10.3	16.7	16.8	13.3	9.8	6.3	2.7
	5%	DB	0.4	0.3	0.5	3.1	7.6	11.8	15.0	16.8	14.1	10.2	6.0	2.2
		MCWB	-0.3	-0.4	-0.2	2.0	5.1	8.8	13.1	15.4	12.6	9.1	5.4	1.3
	10%	DB	-0.3	-0.1	0.1	2.3	6.0	9.6	12.5	15.2	12.9	9.5	5.1	1.1
		MCWB	-1.1	-0.8	-0.6	1.5	4.4	7.7	11.2	13.8	11.3	8.4	4.3	0.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.1	1.4	2.0	4.6	9.9	14.1	20.5	19.0	16.4	11.6	7.5	4.3
		MCDB	1.3	1.5	3.6	7.1	13.9	18.4	21.8	19.8	16.8	12.4	7.9	4.7
	2%	WB	0.4	0.2	0.6	3.0	6.9	10.7	17.2	17.5	14.1	10.2	6.5	2.9
		MCDB	0.7	0.5	0.8	3.3	8.9	13.1	18.3	18.3	15.0	10.8	6.7	3.4
	5%	WB	-0.1	-0.2	0.2	2.3	5.6	9.0	13.3	15.7	12.8	9.3	5.5	1.6
		MCDB	0.3	0.2	0.4	2.8	7.0	11.3	14.7	16.3	13.6	10.0	5.9	2.1
	10%	WB	-1.2	-0.7	-0.3	1.6	4.6	7.7	11.3	14.1	11.7	8.6	4.6	0.3
		MCDB	-0.4	-0.1	0.0	2.1	5.7	9.1	12.3	14.7	12.4	9.3	5.0	1.0
Mean Daily Temperature Range	5% DB	MDBR	3.0	3.1	3.0	2.8	4.2	5.0	4.9	5.3	4.5	3.9	3.4	3.2
		MCDBR	2.9	3.1	3.2	4.5	8.2	9.7	9.4	8.0	6.2	5.2	4.3	4.2
		MCWBR	3.1	3.0	3.3	3.6	5.4	6.6	7.4	6.2	5.2	4.4	4.3	4.6
	5% WB	MCDBR	2.9	3.2	2.8	3.9	7.0	9.2	8.7	7.5	5.7	5.2	4.2	4.3
		MCWBR	3.2	3.3	3.2	3.6	5.3	6.6	7.2	6.6	5.2	4.6	4.5	5.0
Clear-Sky Solar Irradiance	taub	0.358	0.383	0.438	0.486	0.500	0.483	0.468	0.445	0.391	0.401	0.389	0.371	
	taud	2.126	2.081	1.977	1.903	1.936	2.060	2.175	2.245	2.360	2.156	2.091	2.054	
	Ebn at Noon	664	740	757	763	771	789	796	797	810	712	614	578	
	Edh at Noon	89	116	151	179	180	160	141	126	102	106	92	85	
All-Sky Solar Radiation	RadAvg	1.00	1.65	2.73	3.57	3.97	4.18	3.86	3.45	3.19	2.24	1.17	0.81	
	RadStd	0.07	0.15	0.22	0.41	0.53	0.40	0.42	0.28	0.28	0.16	0.09	0.06	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page